

Combinatorial covering designs

OPG-762	L1 - Tractable	Partiellement résolu	Combinatorics
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Énoncé original (anglais, reproduction mot pour mot)

A (v, k, t) covering design, or covering, is a family of k -subsets, called blocks, chosen from a v -set, such that each t -subset is contained in at least one of the blocks. The number of blocks is the covering's size, and the minimum size of such a covering is denoted by $C(v, k, t)$.

Problem Find a closed form, recurrence, or better bounds for $C(v, k, t)$. Find a procedure for constructing minimal coverings.

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